

UNIT 7

INTRODUCTION TO DATABASE SYSTEM



Q.1: DEFINE TERM DATABASE SYSTEM?**DATABASE:**

A database stores data in organized form. Generally, a database is an electronic system that facilitates easy access, manipulation and updating of data. A database is composed of tables which contain rows and columns. These rows and columns are called records and fields respectively. Most databases contain multiple tables.

For example:

A general store database may include tables for purchase, sales and stock records. Each of these tables has different fields that are relevant to the information stored in the table.

Q.2: DEFINE TERMS RECORDS AND FIELDS?

A database is composed of tables which contain rows and columns. These rows and columns are called records and fields respectively. Nowadays, everyone is familiar with the term database; it is a mathematically developed data structure which converts raw input data into meaningful information for a particular organization. These days, a database can be seen in every field of life, for example in industries, health, agriculture, schooling, business and banking. The databases can be developed according to the size of its records for a particular organization.

Databases can be small in size with a few records or very large like NADRA (National Database Registration Authority) databases which keep the multi millions of records.

Advantages of Database:

A database is playing a leading role to enhance the efficiency and performance of any organization. The goal of a database is to minimize the loss and increase the productivity and efficiency of any organization in the age of information technology.

Q.3: DISCUSS THE USE OF DATABASE IN BUSINESS WITH EXAMPLE.

Use of Database in Business:

Businesses use their databases to:

- Keep track of basic transactions
- Provide information that will help the company run the business more efficiently • Help

managers and employees make better decisions

Businesses run on databases. These repositories of organized information can store virtually every kind of data imaginable, and they can sort that information and deliver it to us with a click of a mouse. Business databases help business owners organize and track their customers, inventory and employees.

Example – Customer Relationship Management:

A customer relationship management (CRM) database can help a small business manage the lifeblood of its business its customers. A CRM database organizes all the information a company has about its accounts, contacts, leads and opportunities. A single customer's record may include his contact details, the date and amount of his last order, the total amount of his purchases for the last year, a list of his favorite products and the products he returned, details of customer service calls and more.

Databases can also be used to manage marketing and promotions, to export email addresses and to prepare shipping labels.

Q.4: DESCRIBE DATABASE MANAGEMENT SYSTEM AND NAME SOME POPULAR DBMS.

Database Management System (DBMS):

Databases are usually developed, maintained and controlled by the Database Management System (DBMS). The DBMS essentially serves as an interface between databases and end users or application programs, ensuring that data is consistently organized and remains easily accessible.

Examples:

Here are some examples of popular DBMSs used these days:

MySQL Oracle

- Microsoft SQL Server
- MongoDB
- Visual Foxpro
- IBM Db2
- PostgreSQL

Q.5: WHAT IS THE FLAT FILE SYSTEM?**FLAT FILE SYSTEM:**

Early databases were relatively "flat", which means they were limited to simple rows and columns, like a spreadsheet. A flat file is the older version of database. It stores data in a single table structure. Flat file databases are usually in plain text format, with only one record per line. The fields included in the record are separated using delimiters such as tabs or commas.

Q.6: WHAT IS THE FLAT FILE SYSTEM?**FLAT FILE SYSTEM:**

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Q.7: DESCRIBE THE CHARACTERISTICS OF DATABASE MANAGEMENT SYSTEM.**CHARACTERISTICS OF DATABASE MANAGEMENT SYSTEM:**

A DBMS is modern version of database designing, organization and manipulation. This mainly offers the solutions which a flat file system could not provide. The DBMS has many characteristics. Some of them are:

- (i) Multiple users can access DBMS and can view, add, edit and delete records.
- (ii) A DBMS offers tools like Queries, Views and Forms which help users to manipulate data easily and more efficiently.
- (iii) A DBMS is more secure and reliable.

- (iv) DBMS allows distribution of data in multiple tables by making use of features like keys and relationships between fields of those tables.
- (v) This allows lesser duplication of data and results in lesser redundancy.
- (vi) Preparing backups and providing limited permissions to the users are features of DBMS.
- (vii) DBMS can handle large and complex data more conveniently. Therefore, it is preferred by the medium and large organizations.

Q.8: DEFINE AND DESCRIBE THE BASIC COMPONENTS OF DBMS.**BASIC COMPONENTS OF DBMS:**

The basic components of DBMS are discussed below:

(i) Table:

It is a collection of data elements organized in shape of rows and columns. A contact list may be one of the simplest examples of a table. The marks record prepared by a class teacher is also an example of a table.

(ii) Field:

It is the smallest component in a database. It is where the actual data is stored during data entry. All data field in the same table, have unique names. Fields are also called attributes or columns. Multiple fields make up a data record, several data records make up a data table, and several data tables make up a database.

(iii) Record:

A single entry in a table is called a record. Records are also referred as tuples or rows. A record is made up of two or several data items which are also called tuples in a table representing a set of related data.

For example, the given student table has four tuples/records/rows.

(iv) Data Types:

All fields in a table must have some data type. Data type is a data storage format that can contain a specific type or range of values. The data type of a field is a property that tells what kind of data that field can hold. Here are some basic data types.